



CODEKSHETRA HACKATHON

Round 1: Idea Proposal Document

Track: Software Edition Only

Team Information

No.	Member Name	Institute Name	Role
1	Aditya Dileep Shintre	DKTE	Team Leader
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Team Size Requirement: Minimum 2 and Maximum 4 Members.

Team Name: THE HOISTERS

Project Name: Claim Sheild – Smart Medical Coding and Denial Prediction System

Selected Theme: Health Tech_____

ClaimShield AI – Smart Medical Coding & Denial Prediction System

1. Abstract

Healthcare systems worldwide lose billions annually due to inefficient medical coding and high insurance claim denial rates.

Manual claim processing is slow, error-prone, and reactive, resulting in revenue leakage, delayed reimbursements, and administrative burden.

ClaimShield AI is an AI-powered Revenue Cycle Intelligence Engine that automates medical code extraction from clinical notes and predicts insurance claim denial risk before submission. Using NLP-based diagnosis extraction and machine learning-based denial risk modeling, the system transforms unstructured clinical documentation into structured claim-ready outputs while proactively flagging high-risk claims.

Unlike traditional billing systems that detect errors after claim rejection, ClaimShield AI introduces a predictive pre-adjudication layer that reduces denial probability, improves financial efficiency, and accelerates reimbursements.

2. Problem Statement

Healthcare providers face high insurance claim denial rates (5–20%), leading to revenue loss, increased rework costs, delayed cash flow, and administrative burden. Manual medical coding from unstructured clinical notes is time-consuming and error-prone.

There is a need for an intelligent system that automates clinical-to-code translation, predicts claim denial probability before submission, and assists billing teams in reducing financial risk proactively.

3. Market Analysis – Pros and Cons

Market Need:

- Healthcare administrative waste costs billions annually.
- Increasing insurance policy complexity.
- Rising denial rates due to documentation errors.

Pros:

- Strong demand from hospitals and RCM companies.

- Scalable across healthcare networks.
- High ROI due to reduced denial rework costs.

Cons:

- Requires regulatory compliance in real deployment.
- Integration with existing hospital systems may be complex.
- Model accuracy depends on quality of training data.

4. Target Users

Primary Users:

- Medical Billing Specialists
- Revenue Cycle Management (RCM) Teams
- Hospital Billing Departments

Secondary Users:

- Hospital Revenue Managers
- Healthcare Financial Analysts

5. Existing Solutions

Current solutions include manual coding workflows, static rule-based claim scrubbers, and traditional RCM software with limited AI intelligence.

Limitations:

- Reactive denial handling
- Limited predictive capability
- Lack of intelligent pattern learning

6. Competitive Advantage

- AI-driven medical code extraction from unstructured notes
- Predictive denial risk scoring before claim submission
- Intelligent explanation system for risk factors
- Scalable architecture for integration

7. Market Impact

- Reduce claim denial rates (prototype simulation: 40–60% potential reduction)
- Reduce Accounts Receivable cycle time
- Lower administrative rework costs
- Improve hospital revenue efficiency

8. Proposed Solution

Modules:

1. Clinical NLP Engine – Extracts diagnoses and maps to ICD-10 codes.
2. Claim Structuring Module – Combines ICD, CPT, payer, and charges.
3. Denial Prediction Engine – Uses ML (XGBoost) to predict denial probability and assign risk level.

System Flow:

Clinical Note Upload → Diagnosis Extraction → ICD Mapping → User Adds CPT & Payer → Denial Prediction → Risk Score + Claim Summary Report

9. Technology Stack

Component	Technology Used
Frontend	Streamlit
Backend	FastAPI / Python
Database	SQLite
Cloud / Deployment	Cloud / Localhost
Other Tools / APIs	Scikit-learn, XGBoost, HuggingFace Transformers

Innovativeness

- Predictive denial modeling before submission
- AI-based clinical-to-code automation
- Risk-based claim routing
- Intelligent explanation layer

Show Stoppers and Risk Analysis

Technical Risks:

- Limited dataset for training
- Model accuracy variation

Regulatory Risks:

- Requires compliance in real-world deployment

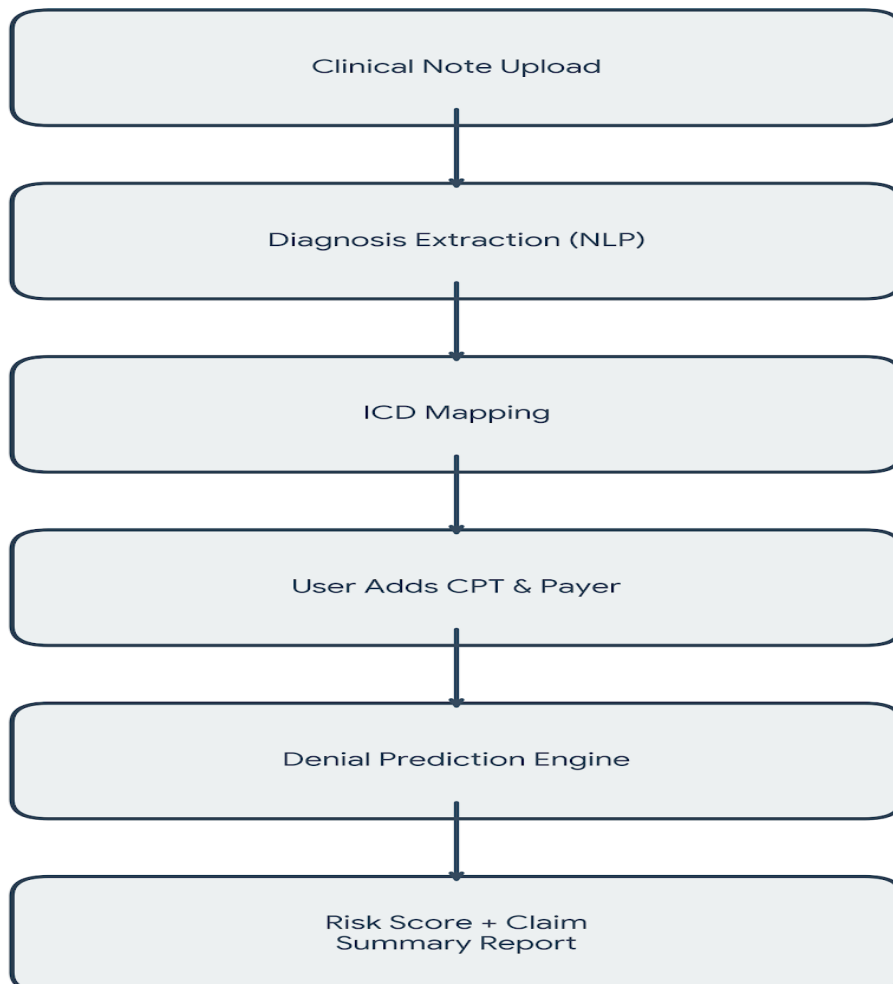
Scalability Challenges:

- Integration with EHR systems
- Multi-payer policy variations

Use Case / System Flow Diagram

System Workflow Diagram:

Workflow Diagram



Prototype / Wireframes

Below are the UI Screenshots of ClaimShield AI:

Upload Clinical Note

ClaimShield AI
Smart Medical Coding & Denial Prediction System

Upload a patient's clinical note, and our AI will extract relevant ICD codes and predict denial risk for insurance claims.

1. Upload Clinical Note

Drop clinical note file here > or click to upload

clinical_note_example.txt

Chief Complaint:
Hypertension and diabetes mellitus follow up.
History of present illness: The patient was admitted with a history of hypertension and type 2 diabetes mellitus.
Blood pressure today 110/80 mmHg.
Patient report consistent high blood pressure reading at home.
Medications: Lisinpril 20mg daily, Metformin 1000mg twice daily.

Assessment:
Hypertension, uncontrolled; Type 2 diabetes mellitus, without complication.
Plan: Adjust Lisinpril to 40mg daily. Continue Metformin at current dose.
Order repeat blood tests including HbA1c and lipid panel.

Process Note

ICD Code Extraction

ClaimShield AI
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2. Extracted ICD Codes

Diagnosis	ICD-10 Code
Essential Hypertension	I10
Type 2 Diabetes Mellitus; Without Complications	E11.9

ICD-10 codes extracted completed successfully.
Please proceed by selecting the CPT code and payer below.

Select CPT Code:
99214 - Office Visit, Level 4

Payer:
Medicare

Predict Denial Risk

Denial Risk Dashboard

Denial Risk Dashboard

ClaimShield AI
Smart Medical Coding & Denial Prediction System

3. Denial Risk Prediction

Low Medium High Critical

Denial Risk: 72% Model Confidence: 85%

Explanation

This claim is classified as HIGH RISK because the combination of ICD 110 Essential Hypertension (with CPT code 99214 (Office Visit, Level 4). Has historicals upon a nly denial rate rate Medicare often, 23) to insufficient medical history necessary documentation.

Claim Details:

Diagnosis:	Essential Hypertension
CPT:	99214 - Office Visit, Level 4
Payer:	Medicare
Estimated Charge:	\$500

Claim Summary Report

Claim Summary Report

ClaimShield AI
Smart Medical Coding & Denial Prediction System

4. Claim Ready Summary

Patient: Demo Patient

Diagnosis: Essential Hypertension
E11.9 - Type 2 Diabetes Mellitus Without Complications

CPT Code: 99214 - Office
Payer: Medicare
Estimated Charge: 3200

Risk Explanation

This claim is classified as HIGH RISK because the combination of ICD 110 Essential Hypertension (with the CPT code 99214 (Office Visit, Level 4). Has historicals upon a nly denial rate rate Medicare often use at insufficient medical history necessary documentation.

Resubmission: And medical reviewers.

Download Claim Report (PDF)

Declaration

We declare that this idea is original and developed for CODEKSHETRA 2026 Hackathon. Any reused components have been properly disclosed.